

Multiple Choice (10 marks)

1. Which of the following does authentication aim to accomplish?
 - (a) Restrict what operations/data the user can access
 - (b) Determine if the user is an attacker
 - (c) Flag the user if he/she misbehaves
 - (d) Determine who the user is
2. _____ is a popular tool used for network analysis in multiprotocol diverse network.
 - (a) Snort
 - (b) SuperScan
 - (c) Burp Suit
 - (d) EtterPeak
3. _____ is a process of wireless traffic analysis that may be helpful for forensic investigations or during troubleshooting any wireless issue.
 - (a) Wireless Traffic Sniffing
 - (b) WiFi Traffic Sniffing
 - (c) Wireless Traffic Checking
 - (d) Wireless Transmission Sniffing
4. _____ is the central node of 802.11 wireless operations.
 - (a) WPA
 - (b) Access Point
 - (c) WAP
 - (d) Access Port
5. _____ is a widespread app's coding mistake made by developers which could be exploited by an attacker for gaining access or malfunctioning your system.
 - (a) Memory leakage
 - (b) Buffer-overflow
 - (c) Less processing power
 - (d) Inefficient programming

True / False (10 marks)

Answer True or False

6. True or False: John the Ripper is primarily used for network intrusion detection and real-time traffic analysis in cybersecurity.
7. True or False: Installing and configuring a firewall is sufficient to detect all instances of IP address spoofing on a network.
8. True or False: A digital signature is based on a public-key system that enables secure authentication and non-repudiation.
9. True or False: Authorization aims to restrict what operations and data a user can access within a system.
10. True or False: WPA 2 is considered to be more secure than WPA 3 due to its longer encryption keys and additional features.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the significance of authentication mechanisms in the presentation layer of cybersecurity and analyze how weaknesses in these mechanisms can impact overall system security.
12. Discuss how blinding techniques, as described in Brumley and Boneh's paper on side-channel attacks, mitigate the effectiveness of timing attacks in cryptographic systems.

End of Paper